

*** DEMO / FICTITIOUS PROPOSAL — FOR TOOL TESTING ONLY ***

TECHNO-COMMERCIAL PROPOSAL

Copper Smelting & Refining Plant — Anode Production Phase 2

Bidder	M/s. Mitsui Mining & Smelting Co. Ltd.
Registered Office	1-11-1 Osaki, Shinagawa-ku, Tokyo 141-8584, Japan
Proposal Ref	MMSC/IN/2026/Q-04471
Date	20 May 2026
Client	Glencore Technology (TCE consultant)
Project	Techno-Commercial Analysis
Tender Enq. No.	TCE/2026/ENQ/4471
Job No.	14734A
Packages bid	TRF, TBRC, TSF

This document is a FICTITIOUS sample created solely to test the TCE Techno-Commercial Evaluator tool. None of the figures or statements herein represent actual offers from the named entity.

1. BID QUALIFICATION

1.1 Net Worth. M/s. Mitsui Mining & Smelting Co. Ltd. confirms a **positive** net worth as per the latest audited annual report for FY 2024-25 (Annexure F-1). Bidder is **QUALIFIED** on this criterion.

1.2 Annual Turnover. Average annual turnover over the last three financial years is **INR 48500 Crore**, meeting the tender threshold. Bidder is **QUALIFIED**.

1.3 Net Profit (PAT). Profit after tax has been **positive** in each of the last three financial years. Bidder is **QUALIFIED**.

2. PROPOSAL COMPLIANCE

2.1 EMD. Earnest Money Deposit of **USD 60,000** submitted via Wire transfer SWIFT MT103. **Compliant**.

2.2 Power of Attorney. Attached (Annexure A2).

2.3 RFQ Acceptance. Accepted with deviations (see Annex E).

2.4 Deviation list. 3 commercial deviation(s) listed in Annexure E.

3. TECHNICAL PROPOSAL

The bidder's technical offering for each package, with all key parameters tabulated below. Where the bidder fully accepts the TCE reference specification, the value matches; deviations are quantified directly in the corresponding cell.

3.1 Tilting Refining Furnace (TRF)

Parameter	Unit	Vendor's Offered Value
Section 1: FURNACE DESIGN		
Number of Furnaces	No.	0.961 No.
Furnace Type	-	Tilting Refining Furnace
Capacity	-	243 TPD/ 7280 TPM
Section 2: PROCESS CYCLE		
Stage-1 Oxidation Time	hrs	4.675 hrs
Stage-2 Reduction Time	hrs	0.316 hrs
Total Reaction Time	hrs	4.935 hrs
Loading / Unloading Time	hrs	Vendor to specify hrs
Total Cycle Time Available	hrs	8.681 hrs
Section 3: OPERATING CONDITIONS		
Fuel Type	-	Natural Gas
Number of Tuyeres	No.	2.026 No.
Tuyere Gas Rate (Instantaneous)	Nm3/h	873.666 Nm3/h
Tuyere Efficiency	%	61.252 %
Operating Temperature Range	-	As per tender
Section 4: STAGE CHEMISTRY		
Stage-1 Copper in Slag	wt%	56.527 wt%
Stage-1 Oxygen in Bath	ppm	4518.585 ppm
Stage-2 Oxygen in Bath	ppm	1469.34 ppm
Arsenic in Anode	ppm	1315.652 ppm
Section 5: MECHANICAL FEATURES		
Furnace Lining & Binding System	-	Backstay & binding system
Freeboard Burners	-	Natural gas fired
Refining Tuyeres	-	Sidewall mounted (compliant)
Charging System	-	Scrap boat & liquid launder
Tilting Mechanism	-	Skimming & pouring by tilting
Section 6: CASTING WHEEL		
Anode Casting Method	-	Casting wheel

Parameter	Unit	Vendor's Offered Value
Mould Cooling	-	Air + Water cooling (compliant)
Mould Wash Type	-	Yotai Coat COAT-NCF
Mould Wash Consumption	kg/t Cu	0.16 kg/t Cu
Section 7: OFF-GAS TREATMENT SYSTEM		
Afterburner / Post Combustion	-	Refractory lined (compliant)
Gas Cooling / Dilution	-	Dilution air
Dedusting System	-	Baghouse
Dust Handling	-	Recycle / bleed (compliant)
Section 8: INTEGRATION		
Feed from TBRC	-	Raw copper
Copper Scrap Charging	-	High % solid scrap
Anode Dispatch	-	ISAKIDD™ Tankhouse (compliant)
Skim & Reject Recycling	-	ISACYCLE™

3.2 Top Blown Rotary Converter (TBRC)

Parameter	Unit	Vendor's Offered Value
Section 1: CONVERTER DESIGN		
Number of Converters	No.	0.982 No.
Converter Type	-	Top Blown Rotary Converter (compliant)
Capacity (heat size)	t	As per tender t
Vessel Diameter	m	To be specified m
Section 2: PROCESS CYCLE		
Smelting Time	hrs	2.478 hrs
Converting Time	hrs	3.075 hrs
Tapping Time	hrs	0.492 hrs
Total Cycle Time	hrs	6.198 hrs
Section 3: OPERATING CONDITIONS		
Lance Gas Type	-	Oxygen-enriched air
Oxygen Enrichment	%	35.571 %
Lance Gas Flow Rate	Nm3/h	4960.1 Nm3/h
Vessel Rotation Speed	rpm	0.512 rpm
Operating Temperature Range	°C	1200-1350 °C
Section 4: STAGE CHEMISTRY		
Matte Grade (Cu in matte)	wt%	65.713 wt%

Parameter	Unit	Vendor's Offered Value
Slag Cu Content (final)	wt%	1.579 wt%
Fe/SiO2 Ratio in Slag	-	1.303
Sulphur in Blister Copper	ppm	203.08 ppm
Section 5: MECHANICAL FEATURES		
Refractory Lining System	-	Mag-chrome basic brick (compliant)
Lance Insertion Mechanism	-	Top-mounted retractable lance
Vessel Tilting Mechanism	-	Hydraulic tilt drive
Charging System	-	Top hatch with feed chute (compliant)
Section 6: TAPPING & SLAG HANDLING		
Tap Hole Design	-	Cu-jacket water-cooled
Slag Pot Handling	-	Ladle transfer car
Matte Transfer	-	Ladle to TRF (compliant)
Section 7: OFF-GAS TREATMENT		
Afterburner / Post Combustion	-	Refractory lined
Gas Cooling	-	Waste heat boiler + dilution air (compliant)
Dedusting System	-	ESP + baghouse
SO2 Off-gas Routing	-	To acid plant
Section 8: INTEGRATION		
Feed Source	-	Concentrate + flux + scrap (compliant)
Downstream Routing	-	Blister → TRF for fire refining
Slag Destination	-	Slag treatment circuit
Dust Recycle	-	Back to TBRC feed (compliant)

3.3 Top Submerged Lance Smelting Furnace (TSF)

Parameter	Unit	Vendor's Offered Value
Section 1: FURNACE DESIGN		
Number of Furnaces	No.	0.989 No.
Furnace Type	-	Top Submerged Lance Smelting
Hearth Area	m2	As per tender m2
Smelting Rate (concentrate)	TPH	51.757 TPH
Section 2: PROCESS CYCLE		
Operation Mode	-	Continuous
Refractory Campaign Life	yr	2.037 yr
Annual Availability	%	88.806 %

Parameter	Unit	Vendor's Offered Value
Section 3: LANCE & OPERATING CONDITIONS		
Lance Type	-	Submerged top-blown (compliant)
Lance Immersion Depth	mm	302.877 mm
Oxygen Enrichment	%	49.937 %
Lance Air Flow	Nm3/h	12210 Nm3/h
Auxiliary Fuel	-	Natural Gas / coal fines
Operating Temperature	°C	1230.492 °C
Section 4: SLAG & MATTE CHEMISTRY		
Target Matte Grade	wt%	56.728 wt%
Final Slag Cu Content	wt%	0.801 wt%
Fe/SiO2 in Slag	-	1.219
Slag Tap Temperature	°C	1190.242 °C
Section 5: MECHANICAL FEATURES		
Refractory Lining	-	Mag-chrome basic brick
Lance Hoist System	-	Hydraulic with positioning control (compliant)
Freeboard Burners	-	Natural gas fired
Concentrate Feed System	-	Belt + weigh feeders to roof port
Section 6: TAPPING SYSTEM		
Matte Taphole Design	-	Water-cooled copper block (compliant)
Slag Taphole Design	-	Water-cooled copper block, separate level
Tap Frequency	/shift	5.731 /shift
Section 7: OFF-GAS TREATMENT		
Waste Heat Boiler	-	Forced circulation, steam generation
Gas Cooling	-	Evaporative + ID fan
Dedusting System	-	ESP, dust recycle (compliant)
SO2 Handling	-	To single contact acid plant
Section 8: INTEGRATION		
Matte Routing	-	Matte ladles to TBRC
Slag Routing	-	Granulation / slag mill circuit (compliant)
Dust Recycle	-	ESP dust back to feed
Acid Plant Coupling	-	Off-gas to H2SO4 plant

3.x TRF Casting Wheel (TRF_CASTING_WHEEL) — *Not bid by this vendor; package excluded from current proposal.*

4. COMMERCIAL PROPOSAL

Overall risk classification (vendor self-assessment): Medium-High. Per-package pricing and terms below.

4.1 Tilting Refining Furnace (TRF)

Item	Vendor's Offer
Total Lump Sum Price	INR 113,750,000 (Lump sum)
Itemised pricing	Compliant — attached as Annex C
Spares & consumables (2-yr)	INR 9,100,000
Delivery terms	FCA Yokohama
Delivery schedule	40 weeks from PO (ex-works)
Payment terms	15% advance, 75% pro-rata, 10% on PAC
Taxes & duties	Indian customs duties & GST extra
Quote validity	90 days from bid due date
Warranty	24 months from PAC
Completion period	16 months from PO
Performance Guarantee	5% PBG valid till warranty + 90 days
Liquidated damages	0.25% per week, capped at 5% (vendor request)
Scope inclusions	Per tender SoW Annex A
Scope exclusions	Casting wheel package, civil works, utility tie-ins, all Indian taxes & duties
Installation & commissioning	Included in scope

4.2 Top Blown Rotary Converter (TBRC)

Item	Vendor's Offer
Total Lump Sum Price	INR 89,180,000 (Lump sum)
Itemised pricing	Compliant — attached as Annex C
Spares & consumables (2-yr)	INR 7,134,400
Delivery terms	FCA Yokohama
Delivery schedule	40 weeks from PO (ex-works)
Payment terms	15% advance, 75% pro-rata, 10% on PAC
Taxes & duties	Indian customs duties & GST extra
Quote validity	90 days from bid due date
Warranty	24 months from PAC
Completion period	16 months from PO

Item	Vendor's Offer
Performance Guarantee	5% PBG valid till warranty + 90 days
Liquidated damages	0.25% per week, capped at 5% (vendor request)
Scope inclusions	Per tender SoW Annex A
Scope exclusions	Casting wheel package, civil works, utility tie-ins, all Indian taxes & duties
Installation & commissioning	Included in scope

4.3 Top Submerged Lance Smelting Furnace (TSF)

Item	Vendor's Offer
Total Lump Sum Price	INR 129,220,000 (Lump sum)
Itemised pricing	Compliant — attached as Annex C
Spares & consumables (2-yr)	INR 10,337,600
Delivery terms	FCA Yokohama
Delivery schedule	40 weeks from PO (ex-works)
Payment terms	15% advance, 75% pro-rata, 10% on PAC
Taxes & duties	Indian customs duties & GST extra
Quote validity	90 days from bid due date
Warranty	24 months from PAC
Completion period	16 months from PO
Performance Guarantee	5% PBG valid till warranty + 90 days
Liquidated damages	0.25% per week, capped at 5% (vendor request)
Scope inclusions	Per tender SoW Annex A
Scope exclusions	Casting wheel package, civil works, utility tie-ins, all Indian taxes & duties
Installation & commissioning	Included in scope

4.4 TRF Casting Wheel (TRF_CASTING_WHEEL)

Not bidding on TRF Casting Wheel.

5. DOCUMENTS SUBMITTED

Status of mandatory documents per tender Annex D:

Code	Document	Status
A	Process Description	Submitted
B	Substance list	Submitted
C	Plant emission list	Submitted
D	Delivery lead time	Submitted
E	Equipment List including noise emissions	Submitted
F	Plant equipment 3D Model	Not submitted (will be furnished pre-PO)
G	Loading Diagrams	Submitted
H	General Arrangement Drawings	Submitted
I	Electrical Consumer List	Submitted
J	I/O Count based on past projects	Not submitted (will be furnished pre-PO)
K	Typical P&ID based on past projects	Not submitted (will be furnished pre-PO)

6. SIGN-OFF

We confirm that the figures, statements and offered terms in this proposal are valid for the validity period stated above. We undertake to honour the commercial verdict — **Acceptable subject to considering deviations** — subject to the deviations listed in Annexure E and the documents pending furnishing per Section 5.

For M/s. Mitsui Mining & Smelting Co. Ltd.

Authorised Signatory

Date